

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

for

MindMantra

A Full-Stack Mental Wellness and Self-Care Platform

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for MindMantra, a web-based mental wellness application. It is designed to help users monitor emotional well-being, practice mindfulness, maintain a private journal, and receive personalized self-care support. This document serves as the common reference for development, UI/UX design, database and API planning, testing, and faculty evaluation.

1.2 Document Conventions

Functional requirements use FR-XX and non-functional requirements use NFR-XX. The word “shall” indicates a mandatory capability. Priority levels are High, Medium, and Low.

1.3 Intended Audience and Reading Suggestions

- Student Developer – implementation and module planning.
- Project Guide / Faculty Evaluator – assessment of scope and feasibility.
- UI/UX Designer – screen and workflow design.
- Backend / Database Developer – API and data-model planning.
- Software Tester – test-case derivation.

1.4 Project Scope

MindMantra will provide a full-stack platform for personal mental wellness and self-care. Users can register and log in, manage profiles, record daily moods, maintain a private journal, practice guided breathing and mindfulness activities, view wellness progress, receive personalized recommendations, and interact with an AI-based wellness assistant. The application is a wellness companion and is not intended to replace professional mental-health diagnosis or treatment.

1.5 References

- IEEE Std 830-1998 – Recommended Practice for Software Requirements Specifications.
- ISO/IEC/IEEE 29148:2018 – Requirements Engineering.
- React.js, Node.js, Express.js and MongoDB documentation.
- MindMantra project concept and frontend requirements.

2. Overall Description

2.1 Product Perspective

MindMantra is a new, self-contained web application following a three-tier architecture: presentation, application/API, and data. The architecture allows the current frontend project to be extended into a complete full-stack application with authentication, persistent data, REST APIs, and optional AI integration.

2.2 Product Features (Summary)

- Registration/login and profile management.
- Daily mood tracking and history.
- Private personal journal.
- Guided breathing, meditation and mindfulness.
- AI-powered wellness assistant.
- Personalized self-care recommendations and affirmations.
- Progress dashboard and basic analytics.
- Reminders and notifications.
- Mental-health support resources.

2.3 User Classes and Characteristics

User Class	Description	Technical Characteristics
User	Primary user of MindMantra for wellness, mood tracking and self-care.	Basic web/mobile literacy.
Administrator	Trusted user for basic content/user management where enabled.	Moderate technical familiarity.

2.4 Operating Environment

Modern browsers on desktop, laptop, tablet and smartphone. Backend: Node.js. Database: MongoDB Atlas or local MongoDB. Internet is required for cloud functionality and optional AI services.

2.5 Design and Implementation Constraints

- Semester-scale academic project.
- Prefer free/open-source technologies.
- Responsive desktop/mobile UI.
- Private wellness and journal data requires access control.
- Third-party AI services may have availability or usage limits.

2.6 Assumptions and Dependencies

Users have basic web literacy; users provide accurate data; internet connectivity is available; AI features depend on the configured provider; notification services depend on the selected third-party service.

3. System Features (Functional Requirements)

Each requirement can be traced to a UI screen and test case.

3.1 Module: User Registration & Authentication

ID	Requirement	Priority
FR-01	The system shall allow a new user to register using name, email and password.	High
FR-02	The system shall validate email format and password strength.	High
FR-03	The system shall authenticate registered users.	High
FR-04	The system shall issue a secure session/token after successful login.	High
FR-05	The system shall allow users to log out securely.	High
FR-06	The system shall provide secure password recovery.	Medium

3.2 Module: User Profile

ID	Requirement	Priority
FR-07	The system shall allow users to view and update their profile.	High
FR-08	The system shall allow users to set wellness preferences and goals.	Medium
FR-09	The system shall prevent unauthorized access to another user's profile data.	High

3.3 Module: Mood Tracking & Emotional Insights

ID	Requirement	Priority
FR-10	The system shall allow a user to record a daily mood.	High
FR-11	The system shall allow an optional note with a mood entry.	Medium
FR-12	The system shall display mood history chronologically or visually.	High
FR-13	The system shall calculate basic mood trends from stored entries.	Medium

3.4 Module: Personal Journal

ID	Requirement	Priority
FR-14	The system shall allow users to create journal entries.	High
FR-15	The system shall allow users to edit and delete their own entries.	High
FR-16	The system shall keep journal entries private to the authenticated owner.	High
FR-17	The system shall store journal date/time information.	High

3.5 Module: Mindfulness & Self-Care

ID	Requirement	Priority
FR-18	The system shall provide guided breathing exercises.	High
FR-19	The system shall provide mindfulness or meditation activities.	High
FR-20	The system shall provide positive affirmations or motivational content.	Medium
FR-21	The system shall track completed wellness activities.	Medium

3.6 Module: AI Wellness Assistant

ID	Requirement	Priority
FR-22	The system shall provide an AI-based conversational wellness assistant.	Medium
FR-23	The system shall use permitted user context for personalization when appropriate.	Medium
FR-24	The assistant shall provide supportive guidance and shall not present itself as a replacement for a qualified professional.	High

ID	Requirement	Priority
FR-25	The system shall provide appropriate support-resource guidance for potentially urgent situations.	High

3.7 Module: Progress Dashboard & Notifications

ID	Requirement	Priority
FR-26	The system shall display a personalized dashboard of recent mood and wellness activity.	High
FR-27	The system shall display basic progress statistics.	Medium
FR-28	The system shall allow wellness reminders to be configured or received.	Medium
FR-29	The system shall display relevant in-app notifications/reminders.	Medium
FR-30	The system shall provide links/information for appropriate support resources.	High

4. External Interface Requirements

4.1 User Interfaces

- Landing/Home Page – introduction, features and authentication actions.
- Authentication – registration, login and password recovery.
- Dashboard – mood summary, activities, reminders and quick actions.
- Mood Tracker – mood selection, notes and history.
- Journal – create, edit, delete and browse private entries.
- Mindfulness – breathing, meditation and affirmations.
- AI Assistant – conversational wellness interface.
- Profile/Settings – account, preferences and notifications.
- Support Resources – professional/emergency support information.
- All screens shall be clean, consistent and responsive.

4.2 Hardware Interfaces

No specialized hardware is required. Standard computers and smartphones with internet access are sufficient.

4.3 Software Interfaces

Component	Suggested Technology
Frontend	React.js, React Router, Axios, Tailwind CSS / Bootstrap
Backend / API	Node.js + Express.js RESTful JSON APIs
Database	MongoDB + Mongoose
Authentication	JWT + bcrypt
AI Service	Gemini/OpenAI API or approved alternative
Version Control	Git + GitHub
Hosting	Vercel/Netlify, Render/Railway, MongoDB Atlas

4.4 Communication Interfaces

Client-server communication shall use HTTPS and RESTful JSON APIs. Protected endpoints shall validate authentication tokens. AI requests shall use the provider's secure API mechanism. Optional email/push notifications shall use a configured third-party service.

5. Use Case Summary

Actor	Use Case	Brief Description
User	Register / Login	Create an account or sign in.
User	Manage Profile	Update profile and wellness preferences.
User	Track Mood	Record and review moods.
User	Write Journal	Create, edit, delete and view private entries.
User	Practice Mindfulness	Perform breathing, meditation and self-care activities.
User	AI Assistant	Interact with supportive AI wellness functionality.
User	View Progress	Review mood trends and activity completion.
User	Manage Reminders	Configure wellness reminders.
User	Support Resources	Access relevant support information.
Admin	Manage Users / Content	Perform enabled administrative functions.

5.1 Primary Flow - Daily Mood Tracking

1) User logs in → 2) Opens Mood Tracker → 3) Selects mood → 4) Optionally adds a note → 5) Submits → 6) System validates and stores the entry → 7) Dashboard updates history/trends → 8) Entry may be used for permitted personalization.

6. Non-Functional Requirements

6.1 Performance Requirements

ID	Requirement	Priority
NFR-01	Core pages should load within 3 seconds under normal demo conditions.	High
NFR-02	Common CRUD operations should respond within 2 seconds under normal load.	Medium
NFR-03	System should support 50-100 concurrent users for a college demo/pilot.	Medium

6.2 Security & Privacy Requirements

- Passwords shall use one-way hashing; plain-text passwords shall never be stored.
- Protected APIs shall require authentication.
- Users shall not access another user's private journal, mood or profile data.
- Deployed communication shall use HTTPS/TLS.
- Inputs shall be validated and sanitized on client and server.
- Sensitive data shall not be unnecessarily exposed to third-party AI services.

6.3 Usability

The interface shall be simple for first-time users, responsive on desktop/tablet/mobile, easy to navigate, and provide clear error/success messages.

6.4 Reliability & Availability

The system shall target 99% availability during evaluation. Mood and journal data shall persist in the database. Failed operations shall return clear errors without corrupting stored data.

6.5 Maintainability & Portability

Backend code shall use modular routes/controllers/models/middleware; frontend shall use reusable components and organized services. The application shall run on common operating systems and modern browsers.

6.6 Scalability

The database and API design shall avoid unnecessary hard-coded assumptions so additional activities, AI providers, notifications and features can be added without a complete redesign.

7. Software Testing Considerations

Test ID	Req.	Scenario	Expected Result
TC-01	FR-01	Register with valid details	Account created successfully.
TC-02	FR-02	Register with invalid email/weak password	Validation prevents registration.
TC-03	FR-10	Submit daily mood	Entry is validated and stored.
TC-04	FR-15	Edit/delete journal entry	Only owner can modify/delete.
TC-05	FR-22	Send AI assistant message	Response or clear service error shown.
TC-06	FR-09	Access another user's profile	Access denied.
TC-07	FR-26	Open dashboard	Latest data is displayed correctly.

Recommended testing types: unit testing for backend logic/validators, API integration testing using Postman or automated tests, and manual UI/UX testing on multiple browsers and a mobile viewport.

8. Proposed System Architecture

MindMantra will use a three-tier full-stack architecture. The React frontend handles presentation and interaction; Node.js/Express provides authentication and REST APIs; MongoDB provides persistent storage. An external AI service may be accessed through the backend.

Presentation Layer

React.js SPA – authentication, dashboard, mood tracker, journal, mindfulness, AI assistant, profile and support resources.

Application Layer

Node.js + Express.js REST API – authentication, authorization, profile management, mood/journal CRUD, activities, reminders and AI integration.

Data Layer

MongoDB with Mongoose – users, profiles, mood entries, journals, activities, progress, reminders and notifications.

8.1 Suggested Technology Stack

Layer	Technology
Frontend	React.js, React Router, Axios, Tailwind CSS / Bootstrap
Backend	Node.js, Express.js, REST APIs
Database	MongoDB with Mongoose
Authentication	JWT, bcrypt
AI Integration	Gemini/OpenAI API or approved AI service
Version Control	Git & GitHub
Testing	Postman / Jest
Deployment	Vercel/Netlify + Render/Railway + MongoDB Atlas

9. Data Model Overview (for Database Planning)

The following entities provide a practical starting point for MongoDB schema design. Fields can be extended during implementation while preserving privacy and ownership rules.

9.1 Core Entities

Entity	Key Attributes
User	_id, name, email, passwordHash, role, createdAt
Profile	_id, userId, wellnessGoals, preferences, createdAt
MoodEntry	_id, userId, mood, intensity, note, recordedAt
JournalEntry	_id, userId, title, content, createdAt, updatedAt
WellnessActivity	_id, title, type, duration, instructions
ActivityLog	_id, userId, activityId, completedAt, duration
AIConversation	_id, userId, messages, createdAt, updatedAt
Reminder	_id, userId, type, scheduledAt, enabled
Notification	_id, userId, message, type, isRead, createdAt

9.2 Key Relationships

- A User has one Profile.
- A User has many Mood Entries.
- A User has many Journal Entries.
- Users complete many Wellness Activities through Activity Logs.
- A User can have multiple AI Conversations.
- A User can configure multiple Reminders and receive multiple Notifications.

10. Future Enhancements (Out of Current Scope)

- Professional counselor/therapist consultation.
- Video/audio support sessions.
- Advanced AI-based mood-pattern analysis.
- Wearable-device integration.
- Native mobile application.
- Push notifications and calendar integration.
- Multilingual support.
- Advanced longitudinal wellness reports.

11. Glossary

Term	Definition
SRS	Software Requirements Specification
API	Application Programming Interface
REST	Representational State Transfer
JWT	JSON Web Token used for authentication
MERN	MongoDB, Express.js, React.js and Node.js
CRUD	Create, Read, Update and Delete
AI	Artificial Intelligence
UI/UX	User Interface / User Experience
HTTPS	HTTP Secure communication using TLS

End of Document – MindMantra SRS v1.0